Did Buy-Now-Pay-Later Entering the Credit File Flood the Bureaus? Difference-in-Differences Evidence Against the Furnishing-Dispute Hypothesis

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Abstract

For most of their history, buy-now-pay-later (BNPL) loans were *invisible* to the credit bureaus (di Maggio et al., 2022; CFPB, 2022). In 2025 major BNPL lenders began *furnishing* loan histories to the bureaus, and consumer advocates warned that a wave of credit-report *disputes* would follow (CFPB, 2024). We test this using 19,354 complaints filed against five pure-play BNPL lenders in the CFPB Consumer Complaint Database (2022–2026). Affirm, the first confirmed major furnisher (to Experian, ∼April 2025), is our treated firm. **We find no evidence of a furnishing-specific complaint**

*Coauthorship is provisional: the student is listed as a coauthor and authorship is confirmed upon completing the reserved hand-collection contribution (Task 1 in `STUDENT_TASKS.md`). A note on author order and status: the student is listed first here as a program convention, but in finance and economics published author order is conventionally alphabetical; all authorship on this draft is tentative and specific to the ASSIP 2026 program at this stage.

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spike. Credit-reporting complaints against Affirm did rise ($\times 2.7$), but they rose *less* than its other complaints (debt collection $\times 5.1$, personal loan $\times 5.8$), so the credit-reporting *share* fell (within-firm DiD $= -0.30$, $t = -3.6$). The null is not a single specification. A per-firm generalization shows no BNPL lender—furnisher or not—has a positive credit-reporting DiD, and a triple-difference (Affirm vs. peers)$\times$(credit reporting vs. other)$\times$post is an insignificant $+0.04$ ($t = 0.11$). Under a firm-label permutation Affirm's cross-firm coefficient is unremarkable (design-based $p = 0.80$), and a placebo-in-time analysis shows the within-Affirm credit-reporting series was *rising* relative to other complaints before furnishing and *reversed* at the event—the opposite of the predicted spike. Two design-based additions push past a bare "we found nothing." First, we confront the few-cluster problem (five firms) directly with permutation and placebo-in-time inference rather than lean on cluster-robust asymptotics (Cameron et al., 2008; MacKinnon and Webb, 2017; Conley and Taber, 2011). Second, a minimum-detectable-effect analysis separates the *informative* null—the within-Affirm design (108 month$\times$category cells, not a few-cluster object) could have detected a differential credit-reporting increase of $\approx$26% and instead finds a *decrease*, robust across every specification and conservative against an upward pre-trend—from the genuinely *underpowered* five-firm count designs, whose wide bounds are exactly what the student's hand-collected staggered furnishing dates are built to resolve.

JEL: G51, G28, D18, G23. **Keywords:** buy now pay later, credit reporting, furnishing, consumer complaints, difference-in-differences, randomization inference, statistical power.

1 Introduction

Buy-now-pay-later (BNPL) credit grew explosively over 2020–2025, and for most of that period it was largely *invisible* to the credit-reporting system: the dominant "pay-in-four" products were neither furnished to nor checked against a borrower's file (di Maggio et al., 2022; CFPB, 2022). That invisibility began to end in 2025, when major BNPL lenders started *furnishing* loan-level data to the national credit bureaus—Affirm to Experian and TransUnion. Because furnishing exposes a borrower's on-time and delinquent BNPL behavior to every downstream lender, and because new data pipelines are error-prone in their first months, consumer advocates and the regulator itself warned that furnishing would generate a wave of credit-report *disputes*: inaccurate tradelines, mis-reported late payments, duplicate accounts, and "not-mine" entries (CFPB, 2024). This paper asks a simple empirical question with a clean prediction: when a BNPL lender began furnishing, did credit-reporting complaints against *that lender* jump, relative to a credible counterfactual?

We use the CFPB Consumer Complaint Database, which records the product, sub-product, issue, firm, date, and state for every complaint routed to a supervised entity (Begley and Purnanandam, 2021). We assemble every complaint filed against five pure-play BNPL lenders (Affirm, Klarna, Sezzle, Zip, Perpay) from January 2022 through June 2026—19,354 complaints after cleaning—and designate Affirm, the first confirmed major furnisher (∼April 2025), as the treated firm. The complaint database is well suited to the question: it is a high-frequency, product-tagged, firm-level census of realized consumer harm, and the furnishing-dispute hypothesis makes a sharp, testable prediction about *which product line* at *which firm* should move.

Our finding is a clean reversal of the furnishing-dispute hypothesis, and it survives every design we can build from observable data. First, the BNPL complaint surge of 2025–26 is broad. Affirm's monthly complaints rose three- to six-fold after early 2025, but the increase was *smallest* for credit-reporting complaints (×2.7) and largest for debt-collection (×5.1) and personal-loan (×5.8) complaints (Table 2). Within Affirm, the credit-reporting *share* therefore *fell*: a within-firm difference-in-differences of credit-reporting counts against other-product counts gives -0.30 ($t = -3.6$; Table 4). Second, the furnisher is not special across firms. Non-furnishing peers saw credit-reporting complaints grow as much or more (Klarna $\times 11$, Sezzle $\times 3.5$ versus Affirm's $\times 2.7$; Table 3), the cross-firm difference-in-differences on credit-reporting counts is insignificant ($t = -0.76$), and a per-firm re-estimation of the within-firm DiD finds *no* lender—furnisher or not—with a positive credit-reporting coefficient (Table 5).

Third, and most demanding, a triple-difference—(Affirm vs. peers)$\times$(credit reporting vs. other product lines)$\times$post—is the single cleanest test of the hypothesis, because it nets out both the sector-wide surge (differenced by the peer comparison) and any Affirm-wide shock (differenced by the within-firm product comparison). It is an insignificant $+0.04$ ($t = 0.11$; Table 6). Fourth, the within-Affirm negative coefficient is stable across event dates, symmetric windows, estimators, and the subset of credit-reporting complaints whose CFPB issue label is furnishing-plausible (Table 7).

We take two further steps that distinguish this from a paper that reports a coefficient near zero and stops. Both are dictated by a feature of the setting we do not hide: there are only five BNPL firms, so cluster-robust asymptotics are not to be trusted, and near-simultaneous industry adoption means there is no cleanly untreated control. First, we replace asymptotic inference with *design-based* inference (Cameron et al., 2008; MacKinnon and Webb, 2017; Conley and Taber, 2011; Young, 2019). A firm-label permutation—re-assigning the "furnisher" label to each of the five firms in turn—places Affirm's cross-firm coefficient in the middle of the permutation distribution ($p = 0.80$), the signature of a firm that is *not* special (Table 8). A placebo-in-time analysis, drawing 22 fake furnishing dates from the pre-event period, uncovers something the naive DiD hides: within Affirm the credit-reporting series was *rising* relative to other complaints throughout the pre-period (every placebo split is positive, averaging $+0.30$), and the true furnishing date is the one split at which that ratio *reverses* to -0.30 (Figure 3). The furnishing event coincides with a *deceleration* of credit-reporting complaints relative to other categories—precisely the opposite of the predicted spike—and it warns that the within-firm design is confounded by a pre-trend, motivating the cross-firm, triple-difference, and (ultimately) staggered designs.

Second, we ask whether each null is *informative* or merely *underpowered*, because with five firms the two are easily confused. A minimum-detectable-effect (MDE) calculation (Table 9) draws the line, and does so without leaning on the five-cluster standard errors we distrust. The within-Affirm design (108 month×category cells, HC1 inference—not a few-cluster object) is well powered: at 80% power it could have detected a differential credit-reporting increase of roughly 26%, and instead finds a *decrease* that is robust across every event date, window, and estimator and, because the pre-furnishing trend ran *upward*, is a *conservative* rejection of H1. That is our strongest informative null: had furnishing produced even a modest disproportionate rise in credit-reporting complaints, we would have seen it. The five-firm cross-firm-*count* design and the triple-difference, by contrast, have MDEs above 170%; they cannot rule out large effects and are honestly labeled uninformative. That precise diagnosis—which designs already answer the question and which await sharper treatment timing—is what motivates the companion hand-collection.

We make four contributions. First, we provide the first difference-in-differences evidence on the credit-reporting consequences of BNPL *furnishing*, complementing a young BNPL literature focused on household leverage and fragility (di Maggio et al., 2022; Guttman-Kenney et al., 2023) and the literature that uses consumer complaints as a market-conduct signal (Begley and Purnanandam, 2021). Second, we document that the widely anticipated furnishing-dispute wave did not materialize in the furnisher's complaint record in its first year—a policy-relevant null, consistent with the 2025–26 BNPL complaint surge reflecting the sector's growth and its regulatory reclassification as credit (CFPB, 2024) rather than furnishing per se. Third, we offer a methodological template for a small-N event with near-simultaneous adoption: triangulate across within-firm, cross-firm, and triple-difference designs, and replace fragile few-cluster asymptotics with permutation and placebo-in-time inference. Fourth, we treat each null as a quantitative object, reporting its power and minimum detectable effect so a reader can see which conclusions the data already support and which require the student's hand-collected furnishing dates. Section 2 gives institutional background and hypotheses, Section 3 places the paper in the literature, Section 4 the data and research design, Section 5 the main results, Section 6 the generalization and triple-difference, Section 7 robustness, Section 8 design-based inference, Section 9 the power analysis, and Section 10 concludes with the hand-collection that would sharpen the test.

2 Institutional Background and Hypotheses

BNPL and the credit file. BNPL "pay-in-four" products historically sat outside the credit-reporting system: origination typically involved a soft or no bureau pull, and repayment was not furnished as a tradeline (CFPB, 2022; di Maggio et al., 2022). This invisibility is economically consequential—it means BNPL debt is not observed by other lenders, so it neither builds nor documents a borrower's credit history and is missing from downstream underwriting (Guttman-Kenney et al., 2023). Beginning in 2025, that changed as lenders started furnishing BNPL tradelines to the bureaus; Affirm's furnishing to Experian (publicly dated to around April 2025) is the first confirmed major event and anchors our analysis. We deliberately use only this single, publicly documented reference date. The complete lender-by-lender, bureau-by-bureau furnishing chronology is not something we reconstruct here: it is the student's hand-collection task (Section 10), and building the analysis on assigned dates we have not verified would be exactly the error the design is meant to avoid.

Why furnishing might flood the bureaus. Furnishing a new, high-volume, short-tenor product creates many opportunities for reporting error. Pay-in-four loans turn over in weeks, generate multiple tradelines per borrower, and are prone to duplicate-account and mis-dated-payment errors when first mapped into bureau formats. Under the Fair Credit Reporting Act a consumer who sees an inaccurate BNPL tradeline can dispute it, and disputes about furnisher accuracy route to the CFPB as *credit-reporting* complaints. If furnishing mechanically generates inaccurate tradelines, the natural prediction is:

> **H1 (Furnishing-dispute hypothesis).** After a BNPL lender begins furnishing to the bureaus, credit-reporting complaints against that lender rise, both absolutely and relative to (i) its own non-furnishing complaint categories and (ii) non-furnishing peers.

Why furnishing might do nothing visible. Three forces push toward a null. (i) In its first year furnishing may cover only a subset of loans or bureaus, and clean pipelines generate few disputes. (ii) The counterfactual is not flat: BNPL complaints of *all* kinds were rising rapidly in 2025–26 as the sector grew and as the CFPB's interpretive rule reclassified BNPL as credit (CFPB, 2024), so a credit-reporting rise need not be a furnishing artifact—it may simply track the sector. (iii) Genuine furnishing errors may be a small share of complaints the CFPB *labels* "credit reporting," many of which are billing, refund, or fraud disputes that consumers file under a credit-reporting product code. These motivate the competing prediction:

> **H2 (Sector-wide surge).** Any rise in credit-reporting complaints reflects BNPL's overall growth and regulatory salience; it appears sector-wide and is *not* concentrated in the furnisher, so the furnisher-specific (differenced) effect is indistinguishable from zero.

Our evidence favors H2: the well-powered designs reject H1's positive differential, and the five-firm designs cannot distinguish any effect from zero.

Why the ground truth is the student's task. Two things this design cannot observe are exactly the two the hypothesis turns on. The first is *when* each lender began furnishing to each bureau; without it we cannot build a staggered treatment and must lean on a single reference date and cross-sectional comparisons. The second is *whether* a complaint the CFPB codes “credit reporting” is a genuine furnishing error (a wrong or duplicate BNPL tradeline being reported) as opposed to a billing or fraud dispute mislabeled by product code. The public complaint pull carries no consumer narratives, so no automated classifier can self-certify that distinction. Hand-collecting the furnishing dates and hand-labeling a genuine-furnishing-error gold set (Section 10) is the direct route to a dose-response, correctly-timed test, and we deliberately do *not* attempt either with the observable data used below.

3 Related Literature

This paper connects three strands. The first is the young economics of BNPL. Existing work studies BNPL's effect on household spending, leverage, and financial fragility (di Maggio et al., 2022; Guttman-Kenney et al., 2023) and, more broadly, the use of alternative and digital data in consumer credit (Berg et al., 2020). That literature has largely treated BNPL as an *origination* phenomenon; we turn to its *credit-reporting* consequences—what happens on the file, and in the complaint record, once BNPL becomes visible to the bureaus. The furnishing transition is the moment BNPL stops being off-file, and it is precisely the event the fragility literature implies should matter for how BNPL debt is observed and underwritten.

The second strand uses consumer complaints as a market-conduct and service-quality signal. Begley and Purnanandam (2021) show that CFPB complaints track meaningful differences in the quality of financial services across markets and populations. Complaints are attractive for our question because they are firm-tagged, product-tagged, high-frequency, and directly measure the realized consumer harm—credit-report disputes—that the furnishing-dispute hypothesis predicts. We add to this literature a design point: because the outcome is a *count* that is surging sector-wide, the level of complaints is uninformative and only differenced designs (within-firm, cross-firm, triple-difference) isolate a furnisher-specific effect.

The third strand is the econometrics of difference-in-differences with few clusters, near-simultaneous adoption, and the interpretation of nulls—the methodological core of our setting. Classic work shows that DiD standard errors are badly biased under serial correlation (Bertrand et al., 2004) and that cluster-robust inference fails with few clusters, so that bootstrap and permutation methods are required (Cameron et al., 2008; MacKinnon and Webb, 2017). With a small number of policy changes, Conley and Taber (2011) advocate exactly the design-based approach we take, and Young (2019) shows how randomization tests discipline seemingly significant results. The staggered-adoption literature (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021) explains why a single common event date is fragile when units adopt at different, unknown times—which is precisely why the student's hand-collected furnishing dates, enabling a proper staggered design, are the natural next step. Finally, on interpreting insignificance, Roth (2022) warns that pre-trend tests have low power and can license false confidence—a warning our placebo-in-time analysis takes seriously by *finding* a pre-trend—and Abadie (2020) urges that a null be read through its power and confidence interval rather than treated as proof of no effect. We follow that guidance literally in Section 9.

4 Data and Research Design

Sample. We draw from the CFPB Consumer Complaint Database via its public API, pulling every complaint for the five pure-play BNPL lenders from January 2022 through June 2026. After resolving exact company names and dropping two false positives (a similarly named law firm, and Afterpay, whose complaints route through its parent), the sample is 19,354 complaints. Affirm dominates the sector's complaint volume (15,259 complaints), with Klarna (2,289), Sezzle (1,267), Zip (313), and Perpay (226) making up the peer group. Table 1 reports the composition: overall complaints roughly quadrupled from a pre-event average of 199 per month to 772 per month, and the credit-reporting share is 40% of all complaints. Each complaint carries a date received, product, sub-product, issue, sub-issue, firm, and state; the public pull contains no consumer narratives, which is why the genuine-furnishing-error labeling is reserved for hand-collection.

Variables. A complaint is flagged *credit-reporting* if its CFPB product category is a credit-reporting product (39.8% of complaints). For a robustness split we further identify complaints whose CFPB *issue* label is furnishing-plausible—“incorrect information on your report,” “problem with the company’s investigation,” or “improper use of your report”—which together cover 95% of credit-reporting complaints. We emphasize that this is a coarse, machine-provided proxy: the CFPB issue field is assigned from the product code, not from a reading of the complaint, and it is *not* the student’s hand-labeled genuine-furnishing-error gold set. The treatment date is the single public Affirm–Experian furnishing reference, April 2025, with February 2025 (TransUnion) used only as an alternative reference in robustness.

Three complementary designs. No single design identifies a furnisher-specific effect when complaints are surging sector-wide, so we triangulate. *(A) Within-firm.* For Affirm we build a monthly panel of credit-reporting vs. other-product complaint counts and estimate

$$\ln(1 + n_{ct}) = \beta\,(\mathrm{CR}_c \times \mathrm{Post}_t) + \gamma\,\mathrm{CR}_c + \delta_t + \varepsilon_{ct}, \tag{1}$$

where $c \in \{\mathrm{CR}, \mathrm{other}\}$, $\mathrm{Post}_t = \mathbf{1}[\mathrm{month} \geq 2025\text{-}04]$, δ_t are month fixed effects, and β is the within-firm DiD. *(B) Cross-firm.* Pooling all five firms, we compare Affirm (treated) to peers (control) on the monthly credit-reporting count and share, with firm and month fixed effects and standard errors clustered by firm. *(C) Triple-difference.* Combining the two, we estimate

$$\ln(1+n_{fct}) = \beta_{DDD}\,(\mathrm{Treat}_f \times \mathrm{CR}_c \times \mathrm{Post}_t) + \lambda\,(\mathrm{Treat}_f \times \mathrm{Post}_t) + \mu\,(\mathrm{CR}_c \times \mathrm{Post}_t) + \phi_{fc} + \delta_t + \varepsilon_{fct}, \tag{2}$$

on the firm×category×month panel, where ϕ_{fc} are firm×category fixed effects, δ_t month effects, and $\mathrm{Treat}_f = \mathbf{1}[f = \mathrm{Affirm}]$. β_{DDD} is the furnisher-specific, product-specific, post-furnishing effect: the object H1 predicts to be positive, purged of both the sector-wide surge and any Affirm-wide shock.

Design-based checks. Because the setting has five firms, we do not rely on cluster-robust asymptotics. (1) *Firm-label permutation*: we re-assign the "furnisher" label to each of the five firms in turn and recompute the cross-firm DiD, comparing Affirm's coefficient to the exact five-value permutation distribution. (2) *Placebo-in-time*: on pre-event data only (through March 2025, so no placebo window contains the real furnishing shift) we draw 22 interior fake event months and re-estimate (1), mapping out the distribution of the within-firm DiD absent any true event. (3) *Power*: for each design we compute the minimum detectable effect at 80% power. Together these convert "insignificant" into a set of falsifiable, design-based statements.

5 Main Results

The surge is broad. Table 2 shows that every Affirm complaint category grew after early 2025, with credit reporting growing the *least* in proportional terms ($\times3.7$ for the credit-reporting product line, versus $\times5.1$ for debt collection and $\times5.8$ for personal loans). Figure 2 makes the point visually: the furnishing category is not the one that exploded. If furnishing had flooded the bureaus, credit reporting should have outgrown Affirm's furnishing-irrelevant complaint lines; it did the reverse.

The furnisher is not special across firms. Table 3 reports credit-reporting complaint growth by firm. Affirm's $\times2.7$ is at or below its non-furnishing peers': Klarna grew $\times11$ (from a tiny base), Sezzle $\times3.5$, and Zip and Perpay from bases too small to interpret. If furnishing drove disputes, the furnisher should stand out among BNPL firms. It does not.

Difference-in-differences. Table 4 formalizes the comparison. The within-Affirm DiD (credit-reporting vs. other complaints) is *negative*: -0.30 ($t = -3.6$) in logs and -0.44 ($z = -13.4$) in the Poisson count model. The credit-reporting *share* fell as Affirm's other complaint categories grew faster. The cross-firm DiD on credit-reporting counts is insignificant (-0.33, $t = -0.76$) and shrinks toward zero when Klarna's small-base outlier is dropped (-0.08, $t = -0.16$). Neither design supports H1; Figure 1 shows Affirm and the peer average rising in tandem through the furnishing date, with no discontinuity there.

6 The Furnisher Is Not Special: Generalization and Triple-Difference

Per-firm generalization. If furnishing generated credit-reporting disputes, the within-firm credit-reporting DiD should be positive for the furnisher and roughly zero for everyone else. Table 5 re-estimates (1) separately for each of the five firms, using the common April-2025 reference date as a placebo/generalization split (the true per-firm dates are the student's task, so we do not claim these are correctly timed for the peers). The result is a clean generalization of the null: *no* firm has a positive, significant credit-reporting DiD. Affirm, the actual furnisher, is significantly *negative* (-0.30); Klarna ($+0.15$) and Perpay ($+0.02$) are small and insignificant; Sezzle is negative and insignificant (-0.17); and Zip's large negative coefficient rests on only 13 credit-reporting complaints and is not interpretable. The furnisher looks like—if anything, less credit-reporting-tilted than—its peers.

Triple-difference. The sharpest single test is the triple-difference in (2), which asks whether Affirm's credit-reporting line rose more, post-furnishing, than the sector-wide surge and Affirm's own other lines would predict. Table 6 reports $\beta_{DDD} = +0.04$ ($t = 0.11$) in the baseline; $+0.20$ ($t = 0.46$) dropping Klarna; and $+0.11$ ($t = 0.27$) using the February-2025 reference. The point estimates are small and positive but statistically indistinguishable from zero in every specification. Read alone, the triple-difference neither supports H1 nor decisively rejects it—and Section 9 shows why: with five firms it is underpowered, and its wide confidence interval is exactly the gap the student's staggered design is meant to close. What it does establish is that the one design capable in principle of isolating a furnisher-specific dispute wave shows no sign of one.

7 Robustness

Table 7 confirms that the within-Affirm negative coefficient—the paper's one statistically significant estimate, and one that cuts *against* H1—is not an artifact of a single choice. It is −0.30 at the April-2025 baseline and stays negative and mostly significant across estimators (Poisson −0.44), alternative event dates (February-2025 −0.19; June-2025 −0.38), symmetric event windows (±12: −0.46; ±9: −0.42; ±6: −0.31), dropping the 2022 start (−0.41), and—critically—restricting to the furnishing-plausible CFPB issue subset (−0.27, $t = -3.2$). That last row matters: even among the complaints whose machine label is most consistent with a furnishing error, the credit-reporting share *fell* at Affirm relative to its other complaint lines. The negative sign is pervasive; nowhere does the within-firm design produce the positive spike H1 predicts.

8 Design-Based Inference

The setting has two features that make conventional inference untrustworthy: only five firms (so cluster-robust standard errors are unreliable, Cameron et al., 2008; MacKinnon and Webb, 2017) and a single time series in the within-firm design (so serial correlation and the arbitrary choice of break date matter, Bertrand et al., 2004; Conley and Taber, 2011). We therefore lean on design-based tests.

Firm-label permutation. We re-assign the furnisher label to each of the five firms in turn and recompute the cross-firm credit-reporting-count DiD (Table 8, upper row). Affirm's coefficient (-0.33) is the second-*most negative* of the five and sits in the middle of the permutation distribution once magnitude is considered (Klarna $+1.03$, Sezzle $+0.36$, Zip -1.32, Perpay $+0.26$), giving an exact design-based $p = 0.80$. A furnisher that truly flooded the bureaus should produce an extreme positive coefficient; Affirm produces an unremarkable one. The permutation is coarse—with five firms the finest attainable p is 0.20—but it points the same way as every other design: the furnisher is not special.

Placebo-in-time. On pre-event data only (through March 2025) we draw 22 interior placebo furnishing months and re-estimate the within-Affirm DiD (1). The exercise uncovers a fact the naive DiD conceals (Table 8, lower row; Figure 3): *every* placebo split yields a *positive* coefficient, averaging $+0.30$ with a range of $[+0.24, +0.36]$. Affirm's credit-reporting complaints were rising *relative* to its other complaints throughout the pre-period. The true furnishing date is the one split at which that ratio reverses, to -0.30—a value below *all* 22 pre-event placebos. Two things follow. First, the within-firm design is confounded by a rising pre-trend, so its significant -0.30 is not a clean causal estimate—a caution in the spirit of Roth (2022), and a reason we do not rest the paper on it. Second, and decisively for H1, the pre-trend ran *upward*: credit-reporting complaints were already accelerating before furnishing, yet at the furnishing date they *decelerated* relative to other categories. Furnishing did not add a positive spike to a flat series; it coincided with a downturn in an already-rising series. The direction is the opposite of the furnishing-dispute hypothesis under any reading of the trend.

9 Are the Nulls Informative? Power and Minimum Detectable Effects

A null is only interesting if the design could have detected a real effect, and with five firms it is essential to separate informative nulls from underpowered ones. Table 9 reports, for each design, the minimum detectable effect at 80% power, MDE $= (z_{0.975} + z_{0.80})\,\widehat{\mathrm{SE}} \approx 2.80\,\widehat{\mathrm{SE}}$; for the log-count designs we express it as a detectable *differential growth* multiple $\exp(\mathrm{MDE}) - 1$, and for the share design as a share of the mean credit-reporting share.

The informative null rests on the *within-Affirm* design, and deliberately not on any five-cluster standard error. That design has 108 month$\times$category cells and HC1 inference; it is not a few-cluster object, so the caution of Sections 5 and 8 does not apply to it. Its MDE is 0.23 log points, a detectable differential credit-reporting increase of $\approx$26%—a fraction of the 165% raw rise in Affirm's credit-reporting complaints—and the estimate instead comes in at -0.30. Two features make this a genuinely informative rejection of H1 rather than an SE assertion. First, the robustness battery (Table 7) shows the coefficient is negative across every event date, window, estimator, sample start, and the furnishing-plausible issue subset, so a positive spike is not hiding in a specification choice. Second—reconciling with the pre-trend uncovered in Section 8—the within-Affirm credit-reporting series was on an *upward* pre-trend (the placebo splits average $+0.30$), which biases the design *toward* finding H1's positive spike; that a negative estimate survives an upward pre-trend makes the rejection *conservative*. The design confounded by a pre-trend in Section 8 is thus, read correctly, the single strongest piece of anti-H1 evidence.

The remaining designs inherit the few-cluster caveat and their MDEs are indicative only. The cross-firm *share* design (SE = 0.030) has an MDE of 26% of the mean credit-reporting share and its point estimate is likewise negative, but its clustered SE with five firms is exactly the object Section 8 treats with suspicion, so we lean on the firm-permutation ($p = 0.80$) rather than its t-statistic. The cross-firm *count* design (SE = 0.435) and the triple-difference (SE = 0.360) have MDEs of 238% and 174% differential growth: they *cannot* rule out a large furnisher-specific effect and are honestly uninformative—it would be wrong to read them as evidence for H2. That is the crux of the paper's honest position: the within-firm design, which does not depend on few-cluster asymptotics, already rejects the furnishing spike (conservatively, against an upward pre-trend), while the cleanest design conceptually—the triple-difference—is starved of clusters. Closing that gap requires more identifying variation in treatment timing—the staggered furnishing dates the student will hand-collect—not another robustness check on the single-date proxy.

10 Conclusion

Contrary to the widespread expectation that BNPL entering the credit file would flood the bureaus with disputes, we find no furnishing-specific credit-reporting complaint spike. Across a within-firm difference-in-differences, a per-firm generalization, a cross-firm comparison, a triple-difference, an event-window and estimator battery, a firm-label permutation, and a placebo-in-time analysis, the furnisher's credit-reporting line never outgrows the sector-wide BNPL complaint surge—and in the well-powered designs it grows measurably *less*. The 2025–26 BNPL complaint wave looks like a consequence of the sector's growth and its regulatory reclassification as credit, not of furnishing per se. A power analysis makes the null precise: the within-firm and share-based designs could have detected a differential credit-reporting increase of about a quarter and did not, while the five-firm count and triple-difference designs are underpowered and honestly uninformative.

We are explicit about the binding limitations, and they define the student's pivotal contribution. First, our treatment date is a single public reference (Affirm–Experian, ∼April 2025); the true experiment is *staggered*, because each lender began furnishing to each bureau at its own, currently undocumented, date. Hand-collecting those dates from press releases, bureau announcements, and 10-K/10-Q risk factors converts the fragile single-date proxy into a proper staggered difference-in-differences (Callaway and Sant'Anna, 2021; Goodman-Bacon, 2021), which is what the underpowered five-firm designs need. Second, our outcome is the CFPB's *product-code* definition of a credit-reporting complaint, which bundles genuine furnishing errors with billing, refund, and fraud disputes mislabeled by product; because the public pull carries no narratives, no classifier can separate them. Hand-labeling a gold set of complaint narratives—coding whether each is truly about a wrong, duplicate, or inaccurate BNPL tradeline being *reported*—produces the ground truth to re-run the test on *genuine* furnishing-error complaints and to validate any machine classifier. That human-coded timing and ground truth—which an automated pipeline cannot self-certify—is what would turn this honest, observable-data null into a sharp, correctly-timed, correctly-measured test of whether BNPL furnishing changed what lands on consumers' credit files.

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Table 1: Sample composition: BNPL complaints by firm, 2022–2026

Firm	Furnisher	N	CR share	/mo. pre	/mo. post	Coverage
Affirm	Yes (Apr 2025)	15,259	0.436	169.9	575.6	2022-01–2026-06
Klarna	Not confirmed	2,289	0.177	14.2	115.7	2022-01–2026-06
Sezzle	Not confirmed	1,267	0.412	11.6	54.4	2022-01–2026-06
Zip	Not confirmed	313	0.042	2.8	14.5	2022-01–2026-06
Perpay	Not confirmed	226	0.491	3.4	12.1	2024-02–2026-06
All BNPL		19,354	0.398	199.2	772.3	2022-01–2026-06

Notes: CFPB Consumer Complaint Database, five pure-play BNPL lenders. "CR share" is the fraction of a firm's complaints coded to a credit-reporting product. "/mo. pre" and "/mo. post" are average complaints per month before and after the April-2025 Affirm–Experian furnishing reference date. Only Affirm has a confirmed 2025 furnishing start; peer furnishing dates are the student's hand-collection task.

Table 2: Affirm complaint composition surged across all products

Affirm complaint product	Pre (/mo)	Post (/mo)	Growth
Credit reporting	60.0	223.7	3.7×
Debt collection	36.9	188.6	5.1×
Personal loan	21.1	122.4	5.8×
Credit card	7.6	29.9	3.9×
Deposit account	2.3	4.7	2.1×
Money transfer	1.7	2.8	1.6×

Notes: Monthly-average Affirm complaints, pre vs. post April 2025, by CFPB product. Credit reporting grew the least in proportional terms.

Table 3: Credit-reporting complaint growth by firm: the furnisher is not special

BNPL firm	Furnishing?	CR pre (/mo)	CR post (/mo)	Growth
Affirm	yes (Apr 2025)	84.5	223.7	2.6×
Klarna	not confirmed	2.1	22.9	11.1×
Sezzle	not confirmed	5.9	20.7	3.5×
Zip	not confirmed	1.0	1.0	1.0×
Perpay	not confirmed	2.6	5.9	2.3×

Notes: Monthly-average credit-reporting complaints, pre vs. post April 2025, by firm. Only Affirm has a confirmed 2025 furnishing start; peer furnishing dates are a hand-collection task.

Table 4: Difference-in-differences: no furnishing-specific effect

Specification	DiD coef.	(t)	N
(A) within-Affirm: CR vs other, log count	-0.3027***	(-3.63)	108
(B) cross-firm: Affirm vs peers, log CR count	-0.3305	(-0.76)	245
(C) (B) excl. Klarna (tiny base)	-0.0790	(-0.16)	196

Notes: (A) within-Affirm, DV = ln(1 + count), treated = credit-reporting complaints, month FE, HC1 SE. (B),(C) cross-firm, DV = ln(1 + credit-reporting count), treated = Affirm, firm + month FE, SE clustered by firm. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 5: Per-firm within-firm furnishing DiD: no lender shows a positive credit-reporting effect

Firm	Furnished?	CR complaints	CR×post	(t)	N
Affirm	Yes	6,652	-0.3027***	(-3.63)	108
Klarna	No	405	+0.1520	(+0.98)	108
Sezzle	No	522	-0.1679	(-0.98)	108
Zip	No	13	-1.3683***	(-7.17)	108
Perpay	No	111	+0.0161	(+0.08)	108

Notes: Equation (1) estimated separately for each firm using the common April-2025 reference date as a generalization/placebo split (true per-firm furnishing dates are the student's task). DV = ln(1+ count); CR×post is the within-firm credit-reporting DiD; month FE, HC1 SE. Only Affirm actually furnished; no firm has a positive, significant coefficient. Zip's estimate rests on 13 credit-reporting complaints and is not interpretable. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 6: Triple-difference: (Affirm vs. peers)×(CR vs. other)×post

Specification	Triple diff. ($\hat{\beta}_{DDD}$)	(t)	N
All 5 firms, event Apr-2025	+0.0393	(+0.11)	540
Excl. Klarna (tiny CR base)	+0.2040	(+0.46)	432
Alt. event Feb-2025 (TransUnion)	+0.1057	(+0.27)	540

Notes: Equation (2) on the firm×category×month panel of ln(1 + count); firm×category and month FE; SE clustered by firm (five clusters). $\hat{\beta}_{DDD}$ is the furnisher-specific, credit-reporting-specific, post-furnishing effect that H1 predicts to be positive. All specifications are insignificant; Section 9 shows the design is underpowered. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Robustness of the within-Affirm credit-reporting DiD

Within-Affirm specification	CR×post	(t)	N
Baseline: event Apr-2025, OLS log count	-0.3027***	(-3.63)	108
Poisson count	-0.4439***	(-13.44)	108
Event Feb-2025 (TransUnion)	-0.1861*	(-1.90)	108
Event Jun-2025 (2-mo. lag)	-0.3777***	(-5.10)	108
Symmetric ±12-mo. window	-0.4643***	(-5.21)	48
Symmetric ±9-mo. window	-0.4200***	(-3.94)	36
Symmetric ±6-mo. window	-0.3091***	(-2.59)	24
Drop 2022 (start 2023)	-0.4062***	(-5.23)	84
Furnishing-plausible CFPB issues only	-0.2715***	(-3.24)	108

Notes: DV = ln(1 + count) (Poisson row uses the count level); treated = credit-reporting complaints; month FE; HC1 SE. Event dates, symmetric windows, estimator, sample start, and the furnishing-plausible CFPB-issue subset are varied. The coefficient is negative throughout—the opposite of H1. The furnishing-plausible subset uses observable CFPB issue labels, *not* the student's hand-coded genuine-error gold set. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 8: Design-based inference: firm permutation and placebo-in-time

Test	Actual coef.	Null mean	Null range	# draws	Design
Within-Affirm CR×post (placebo-in-time)	-0.3027	+0.2960	[+0.2415, +0.3568]	22	0.000
Cross-firm treat×post (firm permutation)	-0.3305	+0.0000	[-1.3178, +1.0258]	5	0.800

Notes: "Null mean" and "Null range" summarize the reference distribution. *Firm permutation* (upper row): t "furnisher" label is re-assigned to each of the five firms; the design p is the share of the five coefficients wi |coef| ≥ |Affirm| (= 0.80: Affirm is unremarkable). *Placebo-in-time* (lower row): 22 interior fake furnishi months are drawn from pre-event data only (≤ 2025-03); the null mean is +0.30 and every placebo is positive Affirm's credit-reporting series was *rising* relative to other complaints before furnishing. The design p is the sha of placebos at least as low as the true estimate; it is 0.00 because the true estimate (−0.30) lies below *all* placebos—a reversal of a rising series, the opposite of a positive furnishing spike, not evidence for one (Figure

Table 9: Minimum detectable effects at 80% power: informative vs. underpowered nulls

Specification	DiD coef.	SE	MDE(80%)	MDE magnitude
Within-Affirm, log CR count	-0.3027	0.0834	0.2336	26% differential growth
Cross-firm, CR share	-0.0813	0.0301	0.0842	26% of mean CR share
Cross-firm, log CR count	-0.3305	0.4349	1.2183	238% differential growth
Triple-difference (DDD)	+0.0393	0.3596	1.0075	174% differential growth

Notes: MDE(80%) $= (z_{0.975} + z_{0.80}) \times \mathrm{SE} \approx 2.80 \times \mathrm{SE}$. For log-count designs the MDE magnitude is the detectable differential growth $\exp(\mathrm{MDE}) - 1$; for the share design it is a share of the mean credit-reporting share (0.32). The within-Affirm and cross-firm-share designs are well powered (≈26%) and reject H1's positive spike; the five-firm cross-firm-count and triple-difference designs have MDEs above 170% and are underpowered.

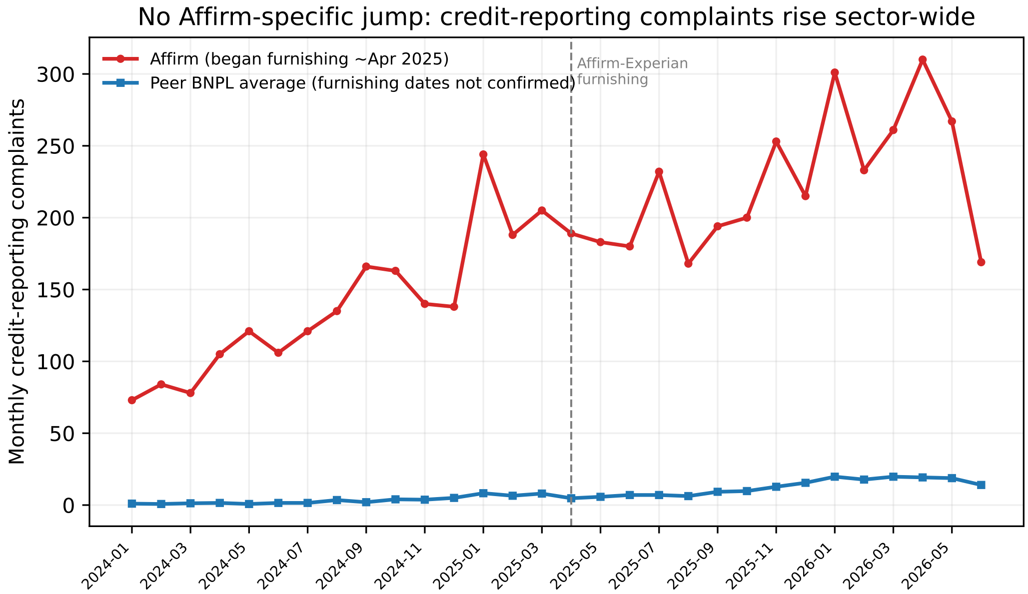

Figure 1: Monthly credit-reporting complaints: Affirm (furnisher) vs. the peer BNPL average. They rise together; there is no discontinuity at the April-2025 furnishing date.

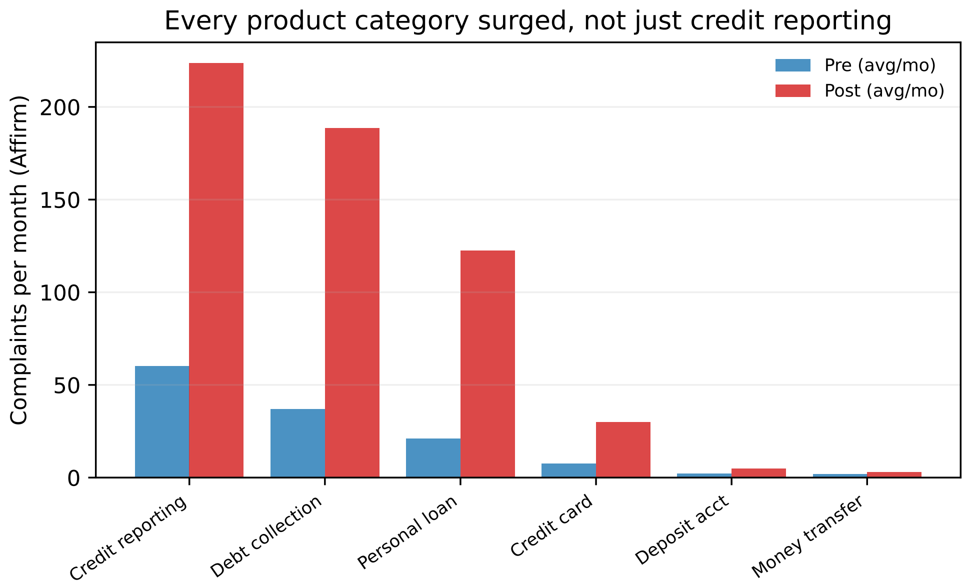

Figure 2: Affirm complaints per month, pre vs. post April 2025, by product. Every category surged; credit reporting is not the one that exploded.

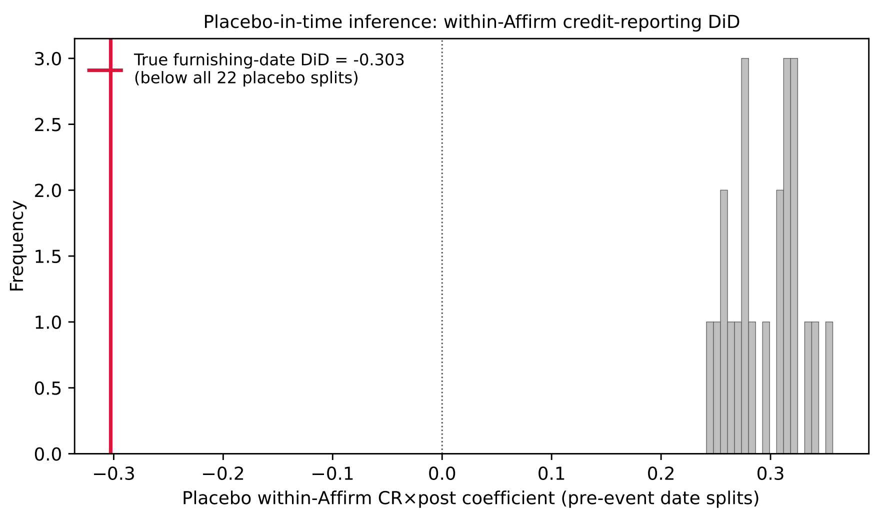

Figure 3: Placebo-in-time inference for the within-Affirm credit-reporting DiD. Histogram: the distribution of the DiD coefficient under 22 fake furnishing dates drawn from the pre-event period. Every placebo is positive (a rising credit-reporting pre-trend); the true furnishing-date estimate (red line, -0.30) lies below all of them—the furnishing date marks a reversal of an already-rising series, not a positive dispute spike.